

Erdős Problem #623

EP-623

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let X be a set of cardinality $\aleph_\omega$ and f be a function from the finite subsets of X to X such that $f(A) \in A$ for all A . Must there exist an infinite $Y \subseteq X$ that is independent - that is, for all finite $B \subset Y$ we have $f(B) \in Y$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>